

- Lab report - Lab 04: Inertial Measurement Unit Sensor

[S26] Sensors and Sensing

- Innopolis University -

Anas Khmais Hamrouni
Vladimir Toporkov
Andrey Krasnov
Askar Kadyrgulov

B23-RO-01

[\[Lab notebook\]](#)

[\[Lab Solution Git Repository\]](#):

**experiments steps/instructions in README*

Topic :

Integration of the MPU6050 IMU Sensor with Raspberry Pi 5 via I2C and ROS2, with Madgwick Filtering and Extended Kalman Filter (EKF) Sensor Fusion

Purpose:

The purpose of this laboratory work is to gain hands-on experience with inertial measurement unit (IMU) sensors:

- Connect the MPU6050 (GY-521 module) to a Raspberry Pi 5 via the I2C protocol and read raw accelerometer and gyroscope data.
- Develop a custom ROS2 Python node that publishes raw IMU data as standard `sensor_msgs/msg/Imu` messages.
- Apply a Madgwick filter to smooth and fuse raw IMU measurements into orientation estimates.
- Feed filtered IMU data into an Extended Kalman Filter (EKF) (`robot_localization` package) for state estimation.
- Visualize and record the system's behavior using RViz2 and rqt_plot.

Description of the sensor application

IMUs are fundamental for robot state estimation. They provide orientation and angular rate data used in navigation, SLAM, balance control, and sensor fusion pipelines.

In this lab, the IMU data pipeline:

MPU6050 (I2C) → ROS2 IMU Node (/imu/raw) → Madgwick Filter (/imu/data) → EKF (`robot_localization`) (/odometry/filtered) → RViz2 visualization

Data conversion to SI units:

- Accelerometer: $\text{accel_m/s}^2 = \text{raw} / 16384.0 \times 9.80665$
- Gyroscope: $\text{gyro_rad/s} = \text{raw} / 131.0 \times (\pi/180)$

List of used equipment/stack:

- **Hardware:** (documentation & datasheets in lab notebook)
 - The MPU6050 is a 6-axis Inertial Measurement Unit. It integrates a 3-axis MEMS accelerometer and a 3-axis MEMS gyroscope on a single chip, communicating via the I2C serial bus.
 - The GY-521 is a breakout board that mounts the MPU6050 chip and exposes VCC, GND, SDA, and SCL pins for easy connection.
 - Raspberry Pi 5

- Dupont Wires
- MicroSD Card
- Power Supply for RPi5
- **Software:**
 - Raspberry Pi OS Imager
 - Docker Engine
 - ROS2
 - `imu_filter_madgwick` ROS2 package
 - `robot_localization` ROS2 package
 - RViz2
 - `rqt_plot`

Description of experiments

Experiment 1 — Raspberry Pi 5 Setup and I2C Sensor Detection (Task 1 & 2)

1. Flashed Raspberry Pi OS (64-bit) onto a microSD card using Raspberry Pi Imager, configuring hostname, SSH, Wi-Fi, and locale.
2. Booted the Raspberry Pi 5, connected via SSH from the laptop.
3. Updated the system (`sudo apt-get full-upgrade`) and installed Docker Engine.
4. Physically wired the GY-521 to the Raspberry Pi 5 GPIO pins (VCC→3V3, GND→GND, SDA→GPIO2, SCL→GPIO3).
5. Enabled I2C interface via `sudo raspi-config` → Interface Options → I2C.
6. Ran `sudo i2cdetect -y 1` to scan the I2C bus and confirm the MPU6050 is detected at address `0x68`.

Experiment 2 — Raw IMU Data Reading Without ROS2 (Task 2 continued)

1. Created a Python test script (`test_imu.py`) on the Raspberry Pi using `smbus2`.
2. The script wakes the MPU6050 by writing `0x00` to register `0x6B`, then continuously reads 14 bytes starting from register `0x3B` (accelerometer X high byte).
3. Parsed the raw bytes into signed 16-bit integers and converted to SI units:
 - Accelerometer: divided by $16384.0 \times 9.80665 \rightarrow \text{m/s}^2$
 - Gyroscope: divided by $131.0 \times (\pi/180) \rightarrow \text{rad/s}$
4. Printed accelerometer [m/s^2] and gyroscope [rad/s] values to the terminal in a loop.

Experiment 3 — Custom ROS2 IMU Publisher Node (Task 3)

1. Created a ROS2 Python package (`ros2 pkg create`) inside `~/ros2_ws/src/` with dependencies `rclpy` and `sensor_msgs`.
2. Implemented `mpu6050_node.py` — a ROS2 node that:
 - Opens the I2C bus using `smbus2`.
 - Wakes the MPU6050 by writing to `PWR_MGMT_1`.
 - Reads 14 bytes from register `0x3B` (accel + temp + gyro) at a configurable rate (default 100 Hz).
 - Converts raw values to m/s^2 and rad/s .
 - Publishes `sensor_msgs/msg/Imu` messages on `/imu/raw` with:

- `header.stamp` and `header.frame_id = "imu_link"`
 - `linear_acceleration.{x,y,z}` in m/s^2
 - `angular_velocity.{x,y,z}` in rad/s
 - `orientation_covariance[0] = -1.0` (orientation not provided)
3. Built the package with `colcon build` and sourced the workspace.
 4. Ran the node and verified with `ros2 topic echo --once /imu/raw` and `ros2 topic hz /imu/raw`.

Experiment 4 — Madgwick Filter for IMU Orientation Estimation (Task 4)

1. Ensured the raw IMU publisher node was running and publishing to `/imu/raw`.
2. Launched the `imu_filter_madgwick` node with remapping and parameters:


```
ros2 run imu_filter_madgwick imu_filter_madgwick_node --ros-args \
  -r imu/data_raw:=/imu/raw \
  -p use_mag:=false \
  -p publish_tf:=false \
  -p publish_debug_topics:=true
```

`use_mag:=false` — magnetometer not available on MPU6050.
`publish_tf:=false` — prevents TF conflicts with the EKF node.
3. Verified outputs:
 - `ros2 topic echo --once /imu/madgwick` — filtered IMU with orientation quaternion.
 - `ros2 topic echo --once /imu/rpy/filtered` — Roll, Pitch, Yaw in radians.

Experiment 5 — EKF Sensor Fusion and RViz2 Visualization (Task 5)

1. Created an EKF configuration file `ekf_imu.yaml` specifying:
 - Input topic: `/imu/data` (output of Madgwick filter)
 - EKF fuses angular velocity (`imu0_config` rows 4: `[true, true, true]` for angular rates)
 - `imu0_remove_gravitational_acceleration: true`
 - Frames: `odom` → `base_link` → `imu_link`
2. Published a static transform: `ros2 run tf2_ros static_transform_publisher 0 0 0 0 0 base_link imu_link`
3. Launched the EKF node: `ros2 run robot_localization ekf_node --ros-args --params-file ekf_imu.yaml`
4. Verified `/odometry/filtered` topic exists and `odom` → `imu_link` TF is live.
5. Opened RViz2, set Fixed Frame to `odom`, added TF and Odometry displays (topic: `/odometry/filtered`).
6. Physically rotated and tilted the IMU sensor while observing motion in RViz2.
7. Tested fixing the IMU with Z-axis pointing up/down to observe yaw drift behavior.
8. Modified `imu0_config` to enable linear acceleration fusion and observed instability/drift.

Description of the results of the experiments:

Result 1 — I2C Detection

Running `sudo i2cdetect -y 1` showed the MPU6050 present at address 0x68 on I2C bus 1, confirming correct wiring and I2C enablement on the Raspberry Pi 5.

```
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file
or directory
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file
or directory
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file
or directory
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file
or directory
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file
or directory
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file
or directory
[carthage@carthage:~$ sudo i2cdetect -y 1
0 1 2 3 4 5 6 7 8 9 a b c d e f
00: -- -- -- -- -- -- -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- 68 -- -- -- -- -- --
70: -- -- -- -- -- -- -- -- -- -- -- -- -- --
[carthage@carthage:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos lab
[carthage@carthage:~$ cd lab
[carthage@carthage:~/lab$ touch test_imu.py
[carthage@carthage:~/lab$ nano test_imu.py
[carthage@carthage:~/lab$ ]
```

Result 2 — Raw Python Script Output

The test script printed accelerometer and gyroscope values continuously. At rest:

```
gyro [rad/s]: 0.06115313944010667 -0.02771209804693287 -0.018785604926045838
accel [m/s^2]: 0.0383072265625 0.9169792358398436 12.57674132080078
gyro [rad/s]: 0.06195252688376819 -0.023848392069235495 -0.021183767257030413
accel [m/s^2]: 0.0143652099609375 0.9600748657226562 12.524068884277343
gyro [rad/s]: 0.05875497710912209 -0.022782542144353463 -0.02331546710679448
accel [m/s^2]: 0.06943184814453125 0.9217676391601562 12.528857287597656
gyro [rad/s]: 0.06035375199644514 -0.02584686067838931 -0.02131699849764067
accel [m/s^2]: 0.15322890625 0.9720458740234375 12.543222497558594
gyro [rad/s]: 0.059554364552783616 -0.02651301688144058 -0.019318529888486854
accel [m/s^2]: 0.21068974609375 1.0510545288085937 12.478579052734375
gyro [rad/s]: 0.05782235842485031 -0.026912710603271342 -0.02424808579106626
accel [m/s^2]: 0.1292868896484375 0.9217676391601562 12.509703674316405
gyro [rad/s]: 0.058888208349732345 -0.023715160828625244 -0.02105053601642016
accel [m/s^2]: 0.040701428222656245 0.9433154541015625 12.478579052734375
gyro [rad/s]: 0.0607534457182759 -0.024781010753507277 -0.020517611053979142
accel [m/s^2]: 0.16519991455078123 0.9935936889648437 12.409147204589843
gyro [rad/s]: 0.05728943346240929 -0.02424808579106626 -0.019984686091538126
accel [m/s^2]: -0.016759411621093748 0.8643067993164062 12.39238779296875
gyro [rad/s]: 0.05742266470301955 -0.024381317031676515 -0.019718223610317616
accel [m/s^2]: -0.05027823486328125 0.9169792358398436 12.445060229492187
gyro [rad/s]: 0.058888208349732345 -0.026644248122050836 -0.02171669221947143
accel [m/s^2]: 0.01915361328125 0.928950244140625 12.524068884277343
gyro [rad/s]: 0.05622358353752726 -0.025313035715940293 -0.022516079663132956
accel [m/s^2]: 0.02154781494140625 0.967257470703125 12.50730947265625
gyro [rad/s]: 0.059154670830952855 -0.024514548272286767 -0.021183767257030413
accel [m/s^2]: 0.14604630126953125 1.024718310546875 12.404358801269531
gyro [rad/s]: 0.057689127184240054 -0.026912710603271342 -0.021849923460081685
accel [m/s^2]: -0.0047884033203125 0.9026140258789062 12.512097875976561
gyro [rad/s]: 0.06115313944010667 -0.0287779479718149 -0.019318529888486854
accel [m/s^2]: 0.0047884033203125 0.9241618408203125 12.516886279296875
gyro [rad/s]: 0.05875497710912209 -0.02904441045303541 -0.023182235866184227
accel [m/s^2]: 0.0023942016015625 0.9528922607421875 12.452242834472656
gyro [rad/s]: 0.05795558966546056 -0.023848392069235495 -0.02198315470069194
accel [m/s^2]: -0.05027823486328125 0.967257470703125 12.490550061035156
gyro [rad/s]: 0.057558959436298 -0.026912710603271342 -0.02198315470069194
```

- Accelerometer Z-axis reads approximately 12.53–12.58 m/s², which is higher than the expected 9.81 m/s² for a flat sensor. It was slightly tilted, causing gravity to project onto multiple axes suggesting a potential scaling offset or calibration error in the MPU6050's accelerometer .
- Accelerometer Y-axis shows ~0.92–0.97 m/s², indicating a small tilt component in that direction.
- Accelerometer X-axis is near zero (~0.01–0.15 m/s²), consistent with minimal tilt in that direction.

- Gyroscope readings show small but non-zero values (~ 0.06 , -0.025 , -0.02 rad/s) despite the sensor being at rest. This bias will cause orientation drift when integrated over time, which is one reason the Madgwick filter and EKF are necessary in subsequent tasks.

Result 3 — ROS2 IMU Topic Verification

```
carthage@carthage:~$ docker exec -it lab-terminal-1 bash
cd ~/ros2_ws
source install/setup.bash
groups: cannot find name for group ID 988
fabian@carthage:~/ros2_ws$ cd ~/ros2_ws
source install/setup.bash
ros2 topic echo --once /imu/raw
header:
  stamp:
    sec: 1771936223
    nanosec: 641275480
  frame_id: imu_link
orientation:
  x: 0.0
  y: 0.0
  z: 0.0
  w: 1.0
orientation_covariance:
- -1.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
angular_velocity:
  x: 0.05542419609386573
  y: -0.025447166956558548
  z: -0.022915773384963714
angular_velocity_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
linear_acceleration:
  x: 0.0909796630859375
  y: 0.9145850341796874
  z: 12.375628381347655
linear_acceleration_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
fabian@carthage:~/ros2_ws$ ros2 topic hz /imu/raw
average rate: 100.023
min: 0.010s max: 0.010s std dev: 0.00003s window: 102
average rate: 100.011
min: 0.010s max: 0.010s std dev: 0.00002s window: 202
average rate: 100.007
min: 0.010s max: 0.011s std dev: 0.00008s window: 302
average rate: 99.756
min: 0.008s max: 0.023s std dev: 0.00064s window: 401
average rate: 99.802
min: 0.008s max: 0.023s std dev: 0.00058s window: 501
average rate: 99.836
min: 0.008s max: 0.023s std dev: 0.00053s window: 602
average rate: 99.859
min: 0.008s max: 0.023s std dev: 0.00049s window: 703
average rate: 99.877
min: 0.008s max: 0.023s std dev: 0.00046s window: 804
average rate: 99.890
min: 0.008s max: 0.023s std dev: 0.00043s window: 904
average rate: 99.802
min: 0.008s max: 0.023s std dev: 0.00052s window: 1004
average rate: 99.820
min: 0.008s max: 0.023s std dev: 0.00050s window: 1105
average rate: 99.835
min: 0.008s max: 0.023s std dev: 0.00047s window: 1206
average rate: 99.848
min: 0.008s max: 0.023s std dev: 0.00046s window: 1307
average rate: 99.859
min: 0.008s max: 0.023s std dev: 0.00044s window: 1407
average rate: 99.868
min: 0.008s max: 0.023s std dev: 0.00042s window: 1508
average rate: 99.877
min: 0.008s max: 0.023s std dev: 0.00041s window: 1609
average rate: 99.884
min: 0.008s max: 0.023s std dev: 0.00040s window: 1710
average rate: 99.890
min: 0.008s max: 0.023s std dev: 0.00039s window: 1810
average rate: 99.896
min: 0.008s max: 0.023s std dev: 0.00038s window: 1910
average rate: 99.901
min: 0.008s max: 0.023s std dev: 0.00037s window: 2010
average rate: 99.858
min: 0.008s max: 0.023s std dev: 0.00042s window: 2109
average rate: 99.865
min: 0.008s max: 0.023s std dev: 0.00041s window: 2210
average rate: 99.810
min: 0.008s max: 0.024s std dev: 0.00050s window: 2310
```

- Rate: The node publishes stably at ~ 100 Hz with minor jitter (~ 0.008 – 0.023 s period) due to OS scheduling and I2C latency.
- Orientation: Set to identity quaternion with `orientation_covariance[0] = -1.0`, correctly signaling to downstream nodes that orientation is not provided.
- Angular velocity: Small non-zero bias at rest (~ 0.055 , -0.025 , -0.023 rad/s), consistent with typical MPU6050 gyroscope zero-rate offset.
- Linear acceleration: Z-axis reads ~ 12.38 m/s² instead of 9.81 m/s², confirming the accelerometer scaling offset observed in Experiment 2.
- Covariances: All set to zero

Result 4 — Madgwick Filter Output

```
anashamrouni — fabian@carthage: ~/ros2_ws — ssh carthage@10.100.2...
tc/apt/sources.list.d/ros2-latest.list:1 and /etc/apt/sources.list.d/ros2.source
s:1
fabian@carthage:~/ros2_ws$ cd ~/ros2_ws
source /opt/ros/humble/setup.bash
source install/setup.bash

ros2 run imu_filter_madgwick imu_filter_madgwick_node --ros-args \
-r imu/data_raw:=/imu/raw \
-p use_mag:=false \
-p publish_tf:=false \
-p publish_debug_topics:=true
[INFO] [1771937003.793479761] [imu_filter_madgwick]: Starting ImuFilter
[INFO] [1771937003.79396228] [imu_filter_madgwick]: Using dt computed from mess
age headers
[INFO] [1771937003.794008766] [imu_filter_madgwick]: The gravity vector is kept
in the IMU message.
[INFO] [1771937003.794175526] [imu_filter_madgwick]: Imu filter gain set to 0.10
0000
[INFO] [1771937003.794249768] [imu_filter_madgwick]: Gyro drift bias set to 0.00
0000
[INFO] [1771937003.794284157] [imu_filter_madgwick]: Magnetometer bias values: 0
.000000 0.000000 0.000000
[INFO] [1771937004.041587769] [imu_filter_madgwick]: First IMU message received.
[]

anashamrouni — colcon build [1/1 done] [0 ongoing] — ssh carthage@10...
ction timed out
[WARN] [1771937108.292012725] [mpu6050_node]: I2C read failed: [Errno 121] Remot
e I/O error
[WARN] [1771937109.303679364] [mpu6050_node]: I2C read failed: [Errno 110] Conne
ction timed out
[WARN] [1771937109.962314984] [mpu6050_node]: I2C read failed: [Errno 121] Remot
e I/O error
[WARN] [1771937110.967626684] [mpu6050_node]: I2C read failed: [Errno 110] Conne
ction timed out
[WARN] [1771937117.322859768] [mpu6050_node]: I2C read failed: [Errno 121] Remot
e I/O error
[WARN] [1771937118.327694753] [mpu6050_node]: I2C read failed: [Errno 110] Conne
ction timed out
[WARN] [1771937121.270170616] [mpu6050_node]: I2C read failed: [Errno 121] Remot
e I/O error
[WARN] [1771937126.391852742] [mpu6050_node]: I2C read failed: [Errno 121] Remot
e I/O error
[WARN] [1771937127.415709436] [mpu6050_node]: I2C read failed: [Errno 110] Conne
ction timed out
[WARN] [1771937136.032497533] [mpu6050_node]: I2C read failed: [Errno 121] Remot
e I/O error
[WARN] [1771937137.047856054] [mpu6050_node]: I2C read failed: [Errno 110] Conne
ction timed out

anashamrouni — fabian@carthage: ~/ros2_ws — ssh carthage@10.100.20.220 — 162x24
WARNING: topic [/imu/madgwick] does not appear to be published yet
Could not determine the type for the passed topic
WARNING: topic [/imu/rpy/filtered] does not appear to be published yet
Could not determine the type for the passed topic
fabian@carthage:~/ros2_ws$ ls
build install log src
fabian@carthage:~/ros2_ws$ cd ~/ros2_ws
source install/setup.bash

ros2 topic echo --once /imu/madgwick
ros2 topic echo --once /imu/rpy/filtered
WARNING: topic [/imu/madgwick] does not appear to be published yet
Could not determine the type for the passed topic
header:
  stamp:
    sec: 1771937051
    nanosec: 851311684
    frame_id: imu_link
  vector:
    x: 0.08186648324636613
    y: -0.010809729667210957
    z: 0.4195290116027524
----
```

- Madgwick node started successfully, receiving IMU messages from `/imu/raw` via the remap `imu/data_raw:=/imu/raw`.
- RPY filtered output: Roll ≈ 0.082 rad ($\sim 4.7^\circ$), Pitch ≈ -0.011 rad ($\sim -0.6^\circ$), Yaw ≈ 0.420 rad ($\sim 24^\circ$), small non-zero angles consistent with the sensor not being perfectly flat and initial gyro bias integration.
- I2C errors on the right terminal (Errno 121 Remote I/O error, Errno 110 Connection timed out) indicate intermittent I2C bus instability, likely caused by loose wiring or insufficient pull-up current, which periodically interrupted the data stream to the Madgwick filter.

Result 5 — EKF Sensor Fusion and RViz2 Visualization ([D link](#))

`ros2 topic list | grep -E "odometry|tf" output:`

`/odometry/filtered`

`/tf`

`/tf_static`

`ros2 topic echo --once`

`/odometry/filtered output:`

header:

stamp:

sec: 1771937753

nanosec: 781156478

frame_id: odom

child_frame_id: base_link

pose:

pose:

position:

x: 0.0

y: 0.0

z: 0.0

orientation:

x: -0.3583370014211173

y: 0.1747377907667483

z: 0.1309935080677269

w: 0.9076904751811536

covariance:

- 1515130.810278492

- 499175.6932319718

...

twist:

twist:

linear:

x: 0.0 y: 0.0 z: 0.0

angular:

x: 0.0574226754044349

y: -0.0247810227927360

z: -0.0222496111623375

covariance:

- 1807.7487798980842

...

```
fabian@carthage:~/ros2_ws$ ros2 topic list | grep -E "odometry|tf"
ros2 topic echo --once /odometry/filtered
ros2 run tf2_ros tf2_echo odom imu_link
/odometry/filtered
/tf
/tf_static
header:
  stamp:
    sec: 1771937753
    nanosec: 781156478
  frame_id: odom
  child_frame_id: base_link
pose:
  pose:
    position:
      x: 0.0
      y: 0.0
      z: 0.0
    orientation:
      x: -0.3583370014211173
      y: 0.1747377907667483
      z: 0.13099350806772697
      w: 0.9076904751811536
  covariance:
    - 1515130.810278492
    - -499175.6932319718
    - -314563.3333800298
    - 0.0
    - 0.0
    - 0.0
    - -499175.6932319688
    - 1052177.3960474983
    - 78113.44163864342
    - 0.0
    - 0.0
    - -314563.3333800277
    - 78113.44163864263
    - 734991.3644751138
    - 0.0
    - 0.0
    - 0.0
    - 0.0
    - 0.0
    - 3.166845006735723
    - 0.33244321032342966
    - 0.9265198325292986
    - 0.0
    - 0.0
    - 0.0
    - 0.33244321032342883
    - 3.1663419412116958
    - 0.5342972811022818
    - 0.0
    - 0.0
    - 0.0
    - 0.9265198325292848
    - 0.5342972811022858
    - 6.177699340890656
  twist:
    twist:
      linear:
        x: 0.0
        y: 0.0
        z: 0.0
      angular:
        x: 0.0574226754044349
        y: -0.024781022792736078
        z: -0.022249611162337504
    covariance:
      - 1807.7487798980842
      - 0.0
```


At time 1771937755.241189032

- Translation: [0.000, 0.000, 0.000]

- Rotation in Quaternion (xyzw):

$[-0.320, 0.157, 0.116, 0.927]$

- Rotation in RPY (radian): [-0.642,

0.374, 0.124]

- Rotation in RPY (degree):

`[-36.755, 21.447, 7.078]`

At time 1771937756.241194516

- Rotation in RPY (degree):

[-33.512, 19.538, 6.924]

At time 1771937757.241192164

- Rotation in RPY (degree):

$[-30.302, 17.625, 6.648]$

[illegible]

- `/odometry/filtered` exists and is publishing, confirming the EKF node is running and fusing IMU data correctly.
- Position is `[0, 0, 0]` since only IMU angular velocity is fused (no position source configured).
- Orientation quaternion is non-trivial (`x: -0.358, y: 0.175, z: 0.131, w: 0.908`), reflecting the actual tilt of the sensor ($\sim 36^\circ$ roll, $\sim 21^\circ$ pitch) consistent with the accelerometer readings seen in previous tasks.
- `odom  $\rightarrow$  imu_link` TF is live and updating every second, with RPY values visibly converging/changing — confirming the full TF tree is functional.
- Twist covariance diagonal values are very large (~ 1807 for angular, ~ 1515130 for pose), indicating the EKF has high uncertainty in position estimates due to zero-covariance inputs from the raw IMU node (all-zero covariance matrices cause numerical instability in the EKF).
- `frame_id: "odom"` was initially missing (tf2_echo reported "Invalid frame ID odom") until the static transform publisher was running, confirming the TF tree dependency.